

AdaptiveSats: Improving Institutional Bitcoin Accumulation Strategies

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1 Introduction

The purpose of this proposal is to outline a strategy for enhancing institutional Bitcoin (Nakamoto (2008)) accumulation. As the demand for Bitcoin continues to grow, it is crucial for institutions to adopt effective methods for acquiring and managing their Bitcoin holdings. This proposal will discuss various approaches and best practices that institutions can implement to optimize their Bitcoin accumulation process.

1.0.1 Problem Statement

Modern markets are dominated by large banks and hedge funds who use sophisticated algorithmic trading strategies as well as market access, and sometimes even non-public information, making it extremely difficult for regular institutions or individuals to compete. Some financial Institutions, particularly smaller sized firms, and retail investors, face a critical wealth accumulation dilemma. Most of them lack the sophisticated resources required to outpace the large banks or gain an edge in the market.

- Bitcoin accumulation represents a social good imperative. It provides a level playing field where information is easily available and is not being controlled by a board of directors, allowing anyone to build strategies for accumulation without offering any undue advantages.
- Bitcoin's public architecture offers transparency and relatively easy access ensuring that critical financial data is available to everyone in the market from the smallest nonprofit to the largest bank simultaneously.
- By leveraging blockchain technology, accumulation can be audited in real-time, eliminating trust barriers.

Dollar-Cost Averaging (DCA) is a common strategy for accumulating Bitcoin, where a fixed amount of money is invested at regular intervals regardless of the asset's price. While DCA is simple and effective for long-term accumulation, it does not take into account market conditions or on-chain signals that could indicate more favorable buying opportunities(Leggio and Lien (2001)).

1.0.2 Objectives

- Identify on-chain/market signals predicting favorable accumulation windows
- Build and validate an adaptive daily allocation model
- Deliver as open-source Python library

2 Data & EDA

- **Source:** BRK `brk_metrics.parquet` (Bitcoin Research Kit Contributors (2025) via Hyper-Trial (2025))
- **Schema:** long format - `day_utc`, `metric`, `value`; zero nulls; all values finite

Property	Value
File size	~1.08 GiB
Total rows	236,259,020
Daily observations	6,274 days
Distinct metric keys	41,407
Date range	2009-01-03 to 2026-03-13

- **Key metric categories to use:** market & valuation (`price_usd`, `mrvv`, `realized_cap`), profitability/SOPR (`adjusted_sopr`, `nupl`, `pain_index`), holder cohorts (`sth_`, `lth_`), supply/scarcity, network activity, technical indicators (`rsi_14d`, `macd_*`)

Figure Placeholder : Non null metric category over time

Figure Placeholder : Bitcoin price history with halving epochs annotated

Figure Placeholder : Metric coverage heatmap by category

Figure Placeholder : Correlation dendrogram of candidate signal features

Figure Placeholder : MVRV / NUPL / SOPR distributions across market cycles

- **Limitations:** requires lazy loading; some metrics start after 2009; extreme multicollinearity across 41k metrics; historical data only.

3 Data Science Approach

3.0.1 Feature Reduction

The dataset contains over 41,000 features, and we aim to identify which features capture similar information to make the dataset more manageable for modeling. To achieve this, we follow a structured feature reduction approach.

First, we group features based on their names using stemming, which helps create feature families such as price-related features. Next, within each family, we apply hierarchical clustering to group features that behave similarly over time, using dendrograms for visualization. Then, from each cluster, we retain a small set of important features and remove those that are highly similar, reducing the dataset size while preserving key information.

This process is computationally intensive when run locally with limited resources, so we may explore more efficient feature reduction methods in future iterations.

3.0.2 Baseline Strategies

To evaluate our approach, we aim to implement a **naive Dollar-Cost Averaging (DCA)** strategy as the baseline, where a fixed amount is invested at regular intervals regardless of market conditions.

This provides a simple benchmark, and our goal is to outperform it by improving accumulation efficiency.

To improve upon this baseline, we are exploring more effective accumulation strategies. In particular, we plan to implement **Value Averaging** as our primary strategy, which adjusts investment based on a target portfolio growth path, allocating more capital during price declines and less during price increases. This aligns well with our objective of maximizing **Sats per Dollar (SPD)**.

In addition, we may also evaluate a **200-day SMA-based strategy**, which adjusts investment based on long-term market trends, providing a simple signal-based benchmark.

3.0.3 Model Development

Our proposed strategy will leverage the reduced feature set to design a data-driven accumulation approach.

The objective is to:

- Identify periods of undervaluation
- Dynamically adjust investment allocation
- Improve accumulation efficiency compared to the defined baseline strategies

3.0.4 Train/Test Split

To avoid look-ahead bias and reflect a realistic backtesting setup, we use a chronological train/test split.

- **Training set:** Data up to December 2023
- **Test set:** January 2024 to March 2026

The test set will remain untouched during model development and will only be used for final evaluation.

3.0.5 Future Work

In future iterations, we may explore additional models and strategies such as LSTM-based models and Hidden Markov Models (HMM). These approaches will be evaluated against the baseline strategies to assess potential improvements in accumulation efficiency.

4 Evaluation & Success Criteria

4.0.1 Metrics

- **Primary Metric (Sats per Dollar - SPD):**
We evaluate accumulation efficiency using Sats per Dollar (SPD), defined as the total Bitcoin accumulated per unit of USD invested. A higher SPD indicates better performance, as it reflects a lower average acquisition cost.
- **Cost Basis Comparison:**
We compare the average USD/BTC acquisition price against the naive DCA baseline over the test period. Lower cost basis indicates improved accumulation efficiency.

- **Smoothness (Allocation Stability):**

We measure the standard deviation of daily allocation weights. Lower variability indicates a more stable investment schedule with potentially lower market impact.

4.0.2 Success Criteria

A strategy will be considered successful if it achieves:

- Higher SPD compared to the naive DCA baseline
- Lower average acquisition cost
- Reduced allocation volatility (more stable investment behavior)
- Consistent performance over the 2024–2026 test window

4.0.3 Stakeholders

- **Trilemma Foundation:** Interested in improving long-term Bitcoin accumulation efficiency
- **Open-source community:** Focused on transparency, reproducibility, and practical usability of the strategy

5 Timeline

Milestone	Target Date
Train/test split + EDA scripts	
Feature reduction + baseline DCA	
ML model + walk-forward CV	
Backtesting + final evaluation	
Final report + library packaging	

References

- Bitcoin Research Kit Contributors. 2025. “Bitcoin Research Kit (BRK).” <https://github.com/bitcoinresearchkit/brk>.
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